MILITARY SPECIFICATION

WELDING MACHINE, ELECTRICAL RESISTANCE, SPOT, ROCKER ARM, SINGLE PHASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical resistance, spot welding machines of the single phase, air or manually actuated, rocker arm design with equipment and accessories specified herein.

1.2 Classification. The welding machine shall be of the following types, classes, and sizes. The type, class, and size, to be furnished shall be as specified (see 6.2.1).

Type I - Air actuated rocker arm.
Type II - Manually actuated rocker arm.

Class A - 8 inch throat depth (for sizes 000,00, and 0).
Class B - 12 inch throat depth (for sizes 000 thru 6).
Class C - 16 inch throat depth (for sizes 000 and 00).
Class D - 18 inch throat depth (for sizes 0 thru 6).
Class E - 24 inch throat depth (for sizes 0 thru 6).
Class F - 30 inch throat depth (for sizes 0 thru 6).
Class G - 36 inch throat depth (for sizes 0 thru 6).
Class H - 42 inch throat depth (for sizes 3.5 thru 6).
Size 000 - 5 KVA at 50 percent duty cycle.
Size 00 - 7.5 KVA at 50 percent duty cycle.
Size 0 - 10 KVA at 50 percent duty cycle.
Size 1 - 15 KVA at 50 percent duty cycle.
Size 2 - 20 KVA at 50 percent duty cycle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Size 3 - 30 KVA at 50 percent duty cycle.
Size 3.5 - 40 KVA at 50 percent duty cycle.
Size 4 - 50 KVA at 50 percent duty cycle.
Size 5 - 75 KVA at 50 percent duty cycle.
Size 6 - 100 KVA at 50 percent duty cycle.

1.2.1 Throat depth. The throat depth is the clear unobstructed distance from the centerline of the electrodes to the first obstruction of the welding machine frame.

1.2.2 Machine size. The machine size is expressed in terms of kilovolt amperes (KVA) at 50 percent duty cycle of the welding transformer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-12323 - Welding, Resistance, Spot, Seam, and Projection for Fabricated Assemblies of Low-carbon Steel.

MIL-W-45210 - Welding, Resistance, Spot, Weldable Aluminum Alloys.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

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U. S. DEPARTMENT OF LABOR

29 CFR, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 5 - Resistance Welding Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA - 56 - Resistance Welding Manual Volume 1.

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new and one of the manufacturer's current models capable of producing commercial quality spot welds in accordance with the requirements herein. The welding machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the welding machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the welding machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

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3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the welding machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, CHAPTER XVII, PART 1910 that are applicable to the welding machine itself. The welding machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial welding machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.8 Environmental protection. The welding machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the welding machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or welding machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the welding machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

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3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

3.4 Components. The welding machine shall consist of not less than a frame and base assembly, enclosure, arm and horn assemblies, electrical system, pneumatic system, and water cooling system.

3.4.1 Machine frame and base. The frame and base assembly of the welding machine shall be a steel weldment of box construction, ribbed to maintain original alignment and to minimize deflection. The base shall provide for floor mounting. Doors and access panels shall be provided with locks and interlocks to prevent access by unauthorized personnel to live portions of the machine. The frame shall be designed to permit positioning of the lower arm and horn assembly.

3.4.2 Enclosure. The enclosure metal shall be corrosion-resistant or treated to resist corrosion. Ventilation and cooling shall be adequate to prevent temperatures in excess of design requirements, under normal operating conditions of rated power output.

3.4.3 Arm and horn assemblies. The arm and horn assemblies shall be designed and constructed to maintain original alignment and to minimize deflection. The horn shall be made of solid copper and have a diameter as specified in table I. Each horn shall permit the electrode holders to rotate and to be adjusted in and out for alignment of the electrodes. The horns shall have clamps that will not mar the electrode holders after repeated adjustment.

3.4.3.1 Rocker arm. The rocker arm shall fulcrum on low-friction bearings and be protected against dust and foreign matter. The type I machine rocker arm movement shall be operated by an air cylinder that is actuated by a foot switch. The type II machine rocker arm movement shall be operated by a manual foot pedal. The rocker arm electrode force shall be adjustable. The rocker arm shall be electrically connected to the secondary winding of the transformer by a flexible shunt having silver plated connections. The electrode openings shall be as shown in table I.

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TABLE I. Machine sizes and dimensions.

Machine size	Welding transformer power rating (KVA)	Electrode opening		Horn diameter	Horn spacing		Electrode holders		
		Air	Manual		Min.	Max.	Length		Taper number
							Upper	Lower	
		(inches)	(inches)	(inches)	(inches)	(inches)	(inches)	(inches)	
000	5	1.25	1.50	1.25	4	8	6	8	RWMA 4 (#1 Morse)
00	7.5	1.25	1.50	1.50	4	8	6	8	RWMA 4 (#1 Morse)
0	10	1.25	1.50	1.50	4	8	6	8	RWMA 4 (#1 Morse)
1	15	1.25	1.50	2.00	5	14	8	12	RWMA 4 (#1 Morse)
2	20	1.25	1.50	2.00	5	14	8	12	RWMA 4 (#1 Morse)
3	30	1.25	1.50	2.50	6	15	8	12	RWMA 4 (#1 Morse)
3.5	40	1.25	1.50	2.50	6	15	8	12	RWMA 5 (#2 Morse)
4	50	1.25	1.50	2.50	6	15	8	12	RWMA 5 (#2 Morse)
5	75	1.25	1.50	2.50	6	15	8	12	RWMA 5 (#2 Morse)
6	100	1.25	1.50	2.50	6	15	8	12	RWMA 5 (#2 Morse)

NOTE: Requirements are "not less than those shown", where a range is shown, the requirements are from the smaller figure or less to the larger figure or more. The electrode opening requirements are based on 12 inch throat depth and are directly proportional to all other throat depths and do not include the requirements for type "I" machines equipped with a retractable stroke.

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3.4.3.2 Lower arm. The lower arm shall be mounted on the frame of the machine to provide a vertical adjustment range of not less than that shown in table I. The arm shall be adjustable by a permanently fastened mechanical device.

3.4.3.3 Electrodes and holders. The electrodes and holders shall conform to table I and to RWMA Bulletin Number 16. Unless otherwise specified (see 6.2.1), one set of electrode holders, of the length specified in table I and one set of straight electrodes of RWMA group A, class 2 alloy with a style F nose shall be supplied.

3.4.4 Electrical system. The electrical system of the welding machine shall include the transformer, rectifiers, timers, meters, suitable circuit protection, disconnect switches, and all controls as required for installation and operation. The electrical system and components shall conform to the applicable requirement of NFPA 70. Unless otherwise specified (see 6.2.1), the machine shall be furnished to operate on single phase, 60 Hz, 230 volts, primary electric service. Each subpanel shall be arranged so that it may be removed independently of any other subpanel. All terminals and units shall be clearly identifiable from the machine electrical wiring diagram. Adequate ventilation of the control cabinets shall be provided. Where ventilation is provided by forced air, the air inlet shall be provided with replaceable filters.

3.4.4.1 Solid state components. Solid state design shall be used throughout the welding machine for electronic components.

3.4.4.2 Main disconnect switch. The machine shall have a main disconnect switch or circuit breaker. The main disconnect switch or circuit breaker shall provide a safety interlock when access doors are opened to areas that are a hazard to the operator. Unless otherwise specified (see 6.2.1), a circuit breaker shall be furnished and shall be mounted on the machine panel. An identified terminal for grounding the welding machine shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 70 for the disconnect fuse rating.

3.4.4.3 Welding transformer. The welding transformer shall be mounted in the frame of the machine. The transformer's KVA rating and secondary current output shall be in accordance with table II for the machine size specified. The transformer for sizes 1 through 6 shall be water cooled and thermostatically controlled to minimize condensation. A temperature responsive device shall be provided to prevent energization of the transformer if the temperature of the transformer exceeds design limits. All static connecting surfaces of the secondary circuit shall be silver plated. The transformer shall be insulated to conform to RWMA Bulletin Number 16 for class B or better insulation. When specified (see 6.2.1), the transformer shall have a series/parallel switch in the primary circuit. Transformer taps connected to a selector switch shall be provided as necessary to produce the required heat range for welding.

3.4.4.4 Control circuit transformer. Control equipment shall be capable of operating on control circuit voltage which varies as much as ±10 percent from the voltage for which the equipment is adjusted. All switch inputs shall

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TABLE II. Machine output capacities.

Machine size	Welding transformer power rating (KVA) 50% duty cycle	Throat depth inches	Horn spacing for maximum ampere test inches	Electrode 1/ force (pounds nominal) manual	Electrode 1/ force (pounds nominal) air	Short circuit current +10 percent amperes	Welding capacities 2/ two equal thicknesses of SAE 1010 steel inches
000	5	8 12 16	6	75	250	7,000 6,000 5,000	.018 .014 .010
00	7.5	8 12 16	6	100	300	8,500 7,500 6,000	.022 .018 .014
0	10	8 12 18 24 36	6	150	350	10,000 9,000 8,500 7,500 6,000	.028 .025 .022 .020 .018
1	15	12 18 24 30 36	8	200	450	10,000 9,000 8,000 7,500 7,000	.031 .028 .022 .021 .020
2	20	12 18 24 30 36	8	300	800	12,000 10,000 9,000 8,500 8,000	.040 .031 .028 .026 .022

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TABLE II. Machine output capacities - Continued.

Machine size	Welding transformer power rating (KVA)	Throat depth	Horn spacing for maximum ampere test	Electrode 1/ force (pounds nominal)		Short circuit current +10 percent	Welding capacities 2/ two equal thicknesses of SAE 1010 steel
	50% duty cycle	inches	inches	manual	air	amperes	inches
3	30	12 18 24 30 36	8	400	1500	19,000 15,500 13,500 12,000 10,500	.090 .064 .057 .045 .036
3.5	40	12 18 24 30 36 42	8	450	1700	22,500 21,000 18,500 17,500 14,500 12,500	.102 .090 .072 .064 .057 .045
4	50	12 18 24 30 36 42	8	500	2000	25,000 21,000 18,000 17,500 14,500 13,500	.114 .102 .072 .020 .057 .045
5	75	12 18 24 30 36 42	8	500	2000	34,000 28,500 25,000 22,000 20,000 18,000	.144 .128 .114 .102 .090 .072

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TABLE II. Machine output capacities - Continued.

Machine size	Welding transformer power rating (KVA)	Throat depth	Horn spacing for maximum ampere test	Electrode 1/ force (pounds nominal)		Short circuit current +10 percent	Welding capacities 2/ two equal thicknesses of SAE 1010 steel
	50% duty cycle	inches	inches	manual	air	amperes	inches
6	100	12	8	500	2000	36,000	.170
	18					34,000	.144
	24					32,000	.128
	30					30,000	.114
	36					28,000	.102
	42					26,500	.090

1/ The pneumatic system being supplied with an air pressure of 80 psi. Maximum value shown are for 12 inch throat depth machines. Electrode forces at other throat depths will vary inversely with the throat depth.

2/ These capacities are for type I machine only. Weld capacities for type II machine will be based on the maximum electrode force of the machine and the manufacturer's recommendations.

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be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage not to exceed 24 volts.

3.4.4.5 Current adjustment control. Continuously variable adjustment of the welding current, from the upper to the lower limits of the machine capacity, shall be provided. The control shall be graduated in increments of 1 from 0 to not less than 99 or from 10 percent to not less than 99 percent for convenience in identifying settings. The means of current adjustment provided shall be such that, at the maximum setting of the adjustment input, the machine maximum short-circuit current produced is not less than the value specified in table II.

3.4.4.6 Type I machine electrical controls. Type I machine electrical controls shall consist of the welding power control, timing and sequence controls, welding contactor, and auxiliary and accessory controls. All calibrations, phase shift controls, and similar characteristics shall be clearly marked in standard terms.

3.4.4.6.1 Operator control panel. The operator control panel shall have an on-off or weld/no weld switch, a power "on" and sequence lights, a current adjustment control, a weld timer selector, auxiliary and accessory control function switches, and any other controls or switches required to meet the requirements specified herein. When furnished, all meters shall have an accuracy of $\pm$2 percent of full scale reading.

3.4.4.6.2 Welding power control. Unless otherwise specified (see 6.2.1), the machine shall be furnished with a synchronous precision combination welding power control unit conforming to, or equivalent to, NEMA ICS 5, type S2H. The control unit may be located in a separate sheet metal cabinet, or mounted as an integral part of the machine.

3.4.4.6.3 Timing and sequencing controls. The control equipment shall provide operating control and adjustment of the welding variables so as to ensure that optimum values of force and current timing and sequencing can be repetitively attained as necessary to produce spot welds in accordance with this specification. Unless otherwise specified (see 6.2.1), the machine shall be furnished with a sequence timer conforming to, or equivalent to, NEMA ICS 5, type 7B. The timing and sequencing functions shall be selected and adjusted by means of knobs, push buttons, or digital switches mounted on the front or side of the machine so as to be easily accessible and in full view of the operator. The timing control input shall be calibrated in cycles (cy.), based on the electrical service line frequency of 60 Hz. The timer shall have at least four functions: squeeze, weld, hold, and off. When specified (see 6.2.1), the controls shall be equipped with functions to perform welds on aluminum alloys in conformance to MIL-W-45210. Additional auxiliary control functions shall be furnished as specified (see 6.2.1). The controls shall have signal lights on the panel to indicate the operating sequence of the welding cycle and to aid in trouble shooting when a malfunction occurs. Use of relays shall be kept to an absolute minimum and only such relays as required to assure adequate safety and reliable operation will be used. Relays shall be the plug-in type where practical. The various timing and sequencing functions shall be grouped on a plug-in chassis. Provisions may be made for

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the removal or addition of extra functions. The control chassis and components shall be mounted in such a manner as to prevent damage to the mounted unit during the service life of the machine.

3.4.4.6.4 Welding contactor. The welding contactor shall be of the silicon controlled rectifier (SCR) type. The SCR contactor shall be mounted in a heat sink and shall be either water or air cooled and thermostatically protected. Air cooled heat sinks may be used on machines up to and including size 3. On water cooled heat sinks, means shall be provided to prevent welding if the water is shut off or if the supply is restricted. The SCR contactor shall be protected from excessive current demands.

3.4.4.6.5 Automatic voltage compensator. When specified (see 6.2.1), the machine shall have an automatic voltage compensator (AVC) system to minimize the effects of line voltage variations. The AVC shall maintain the welding current within ±2 percent for a line voltage variation of ±10 percent. Based on the input line frequency, full compensation shall occur within 1-1/2 Hz for a sudden change in line voltage.

3.4.4.6.6 Accessory controls. Accessory controls shall consist of at least the following.

3.4.4.6.6.1 Foot switch. The foot switch shall control the upper electrode assembly, its associated pneumatic system, and initiate the welding cycle. When specified (see 6.2.1), a two-stage foot switch shall be supplied which shall initiate the electrode force in the first stage and initiate the welding sequence in the second stage. The foot switch shall be guarded to prevent accidental operation of the machine.

3.4.4.6.6.2 Tip dress control. The tip dress control shall operate to lower the upper electrode under a reduced force for cleaning and aligning the electrodes. The control shall provide a pressure and safety limit switch. When applicable, the control may be activated with the first stage of the two-stage foot switch.

3.4.4.6.6.3 Contact pressure gauge switch. When specified (see 6.2.1), the machine shall have a contact pressure gauge switch that shall prevent the initiation of the welding cycle until the predetermined air pressure in the pneumatic head is equal to the contact pressure gauge setting.

3.4.4.6.6.4 Electrode force interlock circuit. The electrode force interlock circuit shall prevent the upper electrode from being raised during the welding cycle. If the retraction switch is released, or a loss of pressure should occur, the weld shall terminate immediately.

3.4.4.6.6.5 Weld/no weld or weld on/off switch. A weld/no weld switch shall be provided. In the no weld or off position, the welding sequence shall be initiated without welding current flow.

3.4.4.6.6.6 Repeat on/off or repeat/non-repeat switch. A repeat on/off switch shall be provided. In the off or non-repeat position only one welding cycle will result with the actuation of the switch. In the on or repeat position, the welding cycle will repeat automatically after a controlled time interval, or off time, as long as the switch remains closed.

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3.4.4.6.6.7 Electrode forge force control. When specified (see 6.2.1), the welding machine shall have an electrode forge force control that shall permit selection of a constant high, constant low, or a variable force sequence at the electrodes for welding.

3.4.4.7 Type II machine electrical controls. The type II machine electrical controls shall consist of a SCR contactor and an electronic timer. The controls shall be operated by a manual foot pedal. When specified (see 6.2.1), the machine shall be equipped with a pass through switch type control in lieu of the SCR contactor. Unless otherwise specified (see 6.2.1), the machine shall be furnished with a timer conforming to, or equivalent to, NEMA ICS 5, type 1A.

3.4.5 Pneumatic system. The pneumatic system and components shall conform to the requirements of RWMA Bulletin Number 16, where applicable, and not in conflict with the requirements of this specification.

3.4.5.1 Pneumatic system operation. Unless otherwise specified (see 6.2.1), the pneumatic system air cylinder shall be a double action, adjustable stroke type, and shall operate with not less then 80 pounds per square inch (psi) input air service. When the adjustable retractable stroke system is specified, the adjustable total stroke shall be not less than 4 inches and shall be either foot or selector switch operated. The inlet connection shall be marked for the pressure requirements. The rocker arm and upper electrode shall operate from a pneumatic system which shall include all necessary valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring, switches, and all related equipment. Air cushions or air speed valves shall be provided to ensure minimum shock operation. The system shall be controlled by a solenoid valve actuated by a foot switch.

3.4.5.2 Air gauges. Air gauges shall have a face diameter of not less than 4-1/2 inches, and shall be calibrated in psi or pounds of applied electrode force. Gauge accuracy shall be within ±0.5 percent of the maximum reading of the gauge. Means of shock protection shall be provided to prevent gauge damage. The gauge shall be mounted on the front or side of the machine and be visible to the operator.

3.4.5.3 Air regulators. The air regulators shall be self-bleeding. Regulators and solenoid valves shall be installed with unions. The regulators shall maintain the pressure at ±1/2 psi within a 2 to 100 psi range, as long as the supply pressure is 2 psi higher than the preset regulator pressure. The air system for the upper electrode shall produce the required electrode force to perform the welds specified herein. If applicable (see 3.4.4.6.6.7), the rise of the electrode force, from its weld force to its forge force level, shall be 75 percent accomplished in no more than 10 milliseconds.

3.4.6 Water cooling system. Unless otherwise specified (see 6.2.1), the electrodes, the SCR contactors on machine sizes 4 through 6, and the transformer on machine sizes 1 through 6 shall be water cooled. The water cooling system, when supplied with water in accordance with the recommendations of the manufacturer, shall operate to prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply, and shall have a single inlet con-

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nection into a manifold. The inlet connection shall be marked for pressure and flow requirements. The system shall have parallel flows from the manifold to the welding transformer, to the secondary winding and circuit conductors, to the primary welding current contactors, and to the upper and lower electrodes. The piping in the system shall be corrosion-resistant to tap water service. The system shall have an inlet strainer, and a closed manifold drain system with flow indicators. The flow indicators may be located in the inlet or outlet side of the water system. The water flow lines to the electrodes shall have solenoid valves to prevent water flow while changing the electrodes. When specified (see 6.2.1), the system shall have water saving devices.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the welding machine shall be not less than those specified in tables I and II for the welding machine size specified.

3.6 Performance. The welding machine shall be capable of producing spot welds conforming to MIL-W-12332 for low carbon steel and, when specified, spot welds conforming to MIL-W-45210 for aluminum alloys, and all additional requirements of tables I and II and this specification.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial welding machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, a metal wrap-around tag, stenciling, silk screening, or by any other method suitable for the use intended. Marking shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each welding machine. If a chart is furnished, it shall be placed in a transparent plastic folder, permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

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Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each welding machine. The nameplate shall contain the information listed below and that information required in RWMA Bulletin Number 16, Name and Rating data plates. If the welding machine is a special model, the model designation shall include the model of the basic standard welding machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's model designation
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The welding machine shall be tested for welding capability in accordance with 3.6. Thirty consecutive welds shall be made using two of the maximum thickness of the material specified in table II for type I machines and the maximum thickness of SAE 1010 steel as recommended by the manufacturers for type II machines. The welds shall comply with the requirements of MIL-W-12332 for low carbon steels. When aluminum alloys welding is specified for type I machines, thirty consecutive welds shall be made using two thicknesses of aluminum alloy of the machines maximum capacity as recommended by the manufacturer. The welds shall comply with the requirements of MIL-W-45210 for aluminum alloy welds

4.7.3 Rated capacity test for the type I machine. The type I machine shall be adjusted to the manufacturer's recommended settings for the maximum thickness of material specified in table II for the size machine being tested. The machine shall be tested under continuous operation for 1 hour, with the two electrodes coming into direct contact, simulating welds at the maximum repetition rate in accordance with RWMA 56 Volume 1, table 2 "Recommended Practice for Single Impulse Spot Welds in Low Carbon Steel." At the end of the test the equipment shall be shut down and all parts examined for over

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heating. The contacts of the relays, if any, shall be examined for evidence that arcing appears to be excessive. If excessive arcing is present, the supplier shall be required to make necessary changes to reduce the arcing. If the equipment fails to conform to any of these requirements, or fails to operate properly during the test, the test shall be discontinued and the supplier shall be permitted to make any necessary adjustments. After adjustments, the test in which the equipment failed shall be completely rerun. Failure to pass the rerun test shall be cause for rejection.

4.7.4 Automatic voltage compensator test. When the AVC is specified (see 3.4.4.6.5), the performance of the AVC shall be measured by means of a calibrated recording instrument. With the current adjustment set first at 50 percent and then at 90 percent of maximum, the secondary current and primary voltage shall be recorded as ±10 percent step changes are introduced into the primary voltage. Alternately, the compensation and response time may be tested by causing the voltage change to occur only on the control of the AVC, in which case, the 10 percent increase or decrease in voltage shall cause, respectively, a decrease or increase of 10 percent in the secondary current within ±2 percent. The secondary current compensation and response time shall be as specified in 3.4.4.6.5.

4.7.5 Dielectric test. The machine shall be subjected to the dielectric test described in RWMA Bulletin Number 16.

4.7.6 Forge force rise time. When the electrode force control is specified (see 3.4.4.6.6.7), the machine sequence shall be set in the variable force mode, and with the machine controls adjusted to the proper values for welding two pieces of the maximum thickness of aluminum alloys as specified in 4.7.2, the rise time of the forge force and the time relationship of the weld current and the forge force shall be measured by means of a calibrated recording instrument. The rise of the electrode force in the variable pressure mode from its weld force level to its forge level shall comply with 3.4.5.3.

4.7.7 Secondary current test. The secondary maximum current shall be determined in accordance with RWMA Bulletin Number 16. The output current shall comply with values specified in table II.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless other specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The welding machines covered by this specification are intended to be used in contractor plants, military, and government repair facilities for general repair and fabrication of sheet metal and wire. They are also used for fabrication of storage boxes and repair of vehicle fenders, air ducts, van skins, and other similar items using weldable aluminum alloys.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition document should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If welding machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Electrodes and holders, if different (see 3.4.3.3).

i. Voltage requirements, if different (see 3.4.4).

j. Circuit breaker and mounting, if different (see 3.4.4.2).

k. Series/parallel switch, if required (see 3.4.4.3).

l. Welding power control, if different (see 3.4.4.6.2).

m. Timing and sequence control, if different (see 3.4.4.6.3).

n. Control functions for welding aluminum alloys, if required (see 3.4.4.6.3).

o. Specify and fully describe additional auxiliary controls, if required (see 3.4.4.6.3).

p. Automatic voltage compensator, if required (see 3.4.4.6.5).

q. Two stage foot switch, if required (see 3.4.4.6.6.1).

r. Contact pressure gauge switch, if required (see 3.4.4.6.6.3).

s. Electrode forge force control, if required (see 3.4.4.6.6.7).

t. Pass through switch type control, if required (see 3.4.4.7).

u. Timer, if different (see 3.4.4.7).

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v. Pneumatic system air cylinder, if different (see 3.4.5.1).

w. Water cooling system, if different (see 3.4.6).

x. Water saving devices, if required (see 3.4.6).

y. Size and capacity, if different (see 3.5).

z. Additional performance features, if required (see 3.6.1).

aa. Standard equipment, if different (see 3.7).

bb. Specify and fully describe optional equipment, if required (see 3.8).

cc. Specify applicable repair parts, if required (see 3.9).

dd. First article inspection, if different (see 4.4).

ee. Quality conformance inspection, if different (see 4.5).

ff. If preservation, packing, and marking in accordance with ML-W-45562 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirement. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The welding machine classification changes from the previous issue of this specification are as follows:

MIL-W-80070C	MIL-W-80070D
Class A	Type I
Class B	Type II
8 inch throat depth	Class A
12 inch throat depth	Class B
16 inch throat depth	Class C
18 inch throat depth	Class D
24 inch throat depth	Class E
30 inch throat depth	Class F
36 inch throat depth	Class G
Size 000	(No change)
Size 00	(No change)

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Size 0	(No change)
Size 1	Added 30 inch throat depth
Size 2	Added 30 inch throat depth
Size 3	(No change)
	Size 3.5 (New size)
Size 4	Added 42 inch throat depth
Size 5	Added 42 inch throat depth
	Size 6 (New size)

6.5 Subject term (key word) listing.

Arm and horn assembly
Compensator
Contact pressure gauge switch
Electrodes
Rocker arm
Timing and sequence controls
Welder, resistance
Welder, rocker arm
Welder, spot
Welding contactor
Welding current
Welding machine, resistance, spot, rocker arm

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - SH
Air Force - 99

Review activity:
Air Force - 84

User activities:
Navy - CG, MC, YD

Preparing activity:
DLA - IP
(Project 3432-0129)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80070D	Welding Machine, Electrical Resistance, Spot, Rocker Arm, Single Phase

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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